Erlang Age-Transport Dynamics: Algebraic Renewal Roots versus the Essential Spectral Edge

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Abstract

We solve an Erlang-fertility McKendrick age-transport semigroup at the level of characteristics, renewal poles, and long-time dynamics. The Euler–Lotka denominator has exactly k explicit algebraic roots. A rank-one resolvent comparison fixes the essential edge at $-\mu$ and proves that a formal root is an L^1 eigenvalue only when it lies strictly to the right of that edge. No renewal pole is promoted to target arithmetic data.

1 Erlang age-transport owner

On $X = L^1(\mathbb{R}_+, da)$, freeze

$$\partial_t n + \partial_a n = -\mu n, \qquad n(t,0) = \int_0^\infty b_k(a) n(t,a)\, da, \tag{1}$$

$$b_k(a) = \frac{\beta\gamma^k a^{k-1} e^{-\gamma a}}{(k-1)!}, \quad \mu, \gamma, \beta > 0, \quad k \geq 1. \tag{2}$$

Age-structured transport descends from M'Kendrick's mathematical population framework [1]; the Erlang specialization and spectral audit are local.

Characteristics transport initial mass to the right with factor $e^{-\mu t}$ and fill the newborn triangle from the boundary. Substitution into (1) produces a scalar renewal equation. Its exponential mode has denominator

$$\Delta(\lambda) = 1 - \beta \left(\frac{\gamma}{\gamma + \lambda + \mu} \right)^k. \tag{3}$$

Therefore the algebraic roots are

$$\lambda_j = \gamma\beta^{1/k} e^{2\pi i j/k} - \gamma - \mu, \qquad 0 \leq j < k. \tag{4}$$

2 Resolvent and the eigenvalue gate

Let $Gf = -f' - \mu f$ with the renewal boundary from (1). For $\Re\lambda > -\mu$, direct integration gives

$$u(a) = e^{-(\lambda+\mu)a} u(0) + \int_0^a e^{-(\lambda+\mu)(a-s)} f(s)\, ds. \tag{5}$$

The boundary determines $u(0)$ through (3). Relative to the absorbing mortality shift, the remaining term in (5) is one scalar functional times one exponential; the resolvent difference is rank one. The essential spectral edge is consequently $\Re\lambda = -\mu$.

Theorem 1 (root/eigenvalue firewall)**.** *A root in (4) is an L^1 eigenvalue exactly when $\Re\lambda_j > -\mu$. Roots on or below the essential edge are not L^1 eigenvalues.*

Proof. Every candidate eigenfunction is a multiple of $e^{-(\lambda_j+\mu)a}$. Its absolute integral is finite exactly under the strict displayed inequality. Equation (3) then supplies, and only then supplies, the renewal boundary. □

References

[1] A. G. M'Kendrick, *Applications of Mathematics to Medical Problems*, Proceedings of the Edinburgh Mathematical Society 44 (1925), 98–130, doi:10.1017/S0013091500034428.